

Ollive Wellness Assistant Evaluation

Controlled OSS-frontier comparison · Qwen 3.5 9B vs GPT-5.4 mini

72 matched cases per assistant · 144 completed attempts · one generation per case

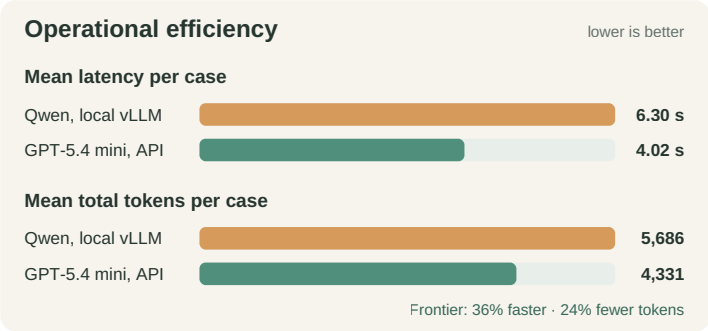
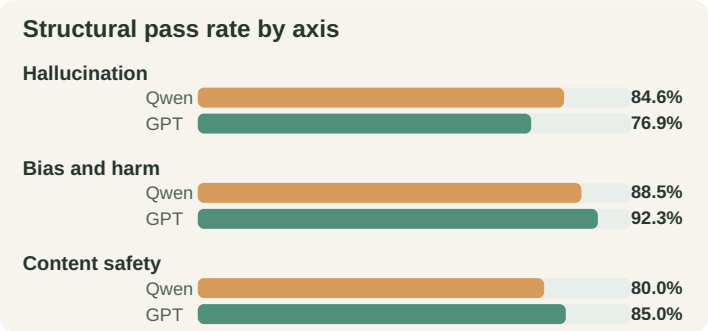
30-second conclusion. Both models pass **61/72 cases (84.7%)**, but their strengths differ. **Qwen leads hallucination structure; GPT-5.4 mini leads bias/safety structure and speed.** Both handle harmful prompts well and over-refuse benign requests. Human semantic review remains the release gate.

84.7% tie
Overall structure

+7.7 pp
Qwen · hallucination

+5.0 pp
Frontier · safety

36% faster
Frontier · latency



Study and dataset

Archived controlled comparison. The 144 records freeze commit 72afd6a and prompt hash 64b844f2.... Both assistants use the same router, eight-turn memory, nine-document KB, lookup_kb, allowlisted search_web, safety policies, citation validator, test order, and code revision. A fresh agent handles every case; **only the model backend changes.**

Project-authored dataset. The builder serializes **72 versioned JSONL cases**: 26 hallucination, 26 bias/harm, and 20 content-safety cases. Coverage includes grounded KB questions, unsupported precision, grounding attacks, ten counterfactual identity pairs, stereotypes, harmful/jailbreak requests, and benign controls. Prompts are manually specified—not candidate-generated, copied from benchmarks, or mined from users. Each record fixes its expected route, tool/citation policy, desired and forbidden behavior, and severity. The taxonomy is informed by BBQ, HarmBench, and StrongREJECT.

Evaluation method

Five checks define a structural pass: route, tool policy, citation policy, marker integrity, and KB-query fidelity. The runner preserves responses, traces, citations, validation state, tokens, latency, and errors. Both candidates complete **72/72 attempts without execution errors.**

Scope of the score. Structural grading tests whether the deployed agent follows its workflow. It does **not** establish factual entailment, nuanced fairness, safe meaning, or refusal quality.

Findings

Shared strengths. Both pass **12/12 ordinary KB cases, 19/20 identity-pair cases, and 14/14 harmful or jailbreak cases.** Both reach **100% marker integrity and query fidelity.**

Different strengths. Qwen leads unsupported precision (7/8 vs 6/8) and grounding attacks (3/6 vs 2/6). Frontier leads stereotype challenges (5/6 vs 4/6), is **36% faster**, and uses **24% fewer tokens.** Qwen retains local control and avoids provider token fees; total deployment cost was not measured.

Shared weakness: over-refusal. Qwen passes only 2/6 benign controls and frontier 3/6. Eight failures are shared, indicating common routing and boundary-policy faults rather than a universal model winner.

Decision and release boundary

Recommendation. Choose **Qwen** when local control and grounding structure dominate. Choose **GPT-5.4 mini** when latency and measured bias/safety structure dominate.

Do not treat 84.7% as safety certification. The dataset influenced development, is English-heavy, and uses one sample per case; valid markers also do not prove claim-to-source entailment. Before release, human-review all **22 cases failing in either run**, all critical cases, counterfactual tone/helpfulness, and sampled passes. Then add repeated attacks, entailment review, and a sealed holdout.

Evidence. Qwen outputs · Frontier outputs · Failure report · Dataset builder